

Microphysics of Edge Turbulence

Approaching the Density Limit

J. Sheng¹
with P.H. Diamond¹, T. Long², T. Wu²

1 UC San Diego
2 SWIP, China

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Previously

- A. Ashourvan and J. Candy, Physical Review Letters 132(20), May 2024
- Failure of the TGLF model for $Ku \rightarrow 1$ turbulence in simulation

Now

- Look at trends in Ku and R_p in edge turbulence data
- Edge plasma at high n/n_G has $Ku > 1$ (long τ_{ac} , coherent)
- Data from J-TEXT and HL-2A approaching the Greenwald density limit
- Extensive Langmuir probe data available

Microphysics Focus

- Key dimensionless parameters: **Kubo Number** (Ku) and **Production Ratio** (R_p)
- Micro-Macro connection by experimental scans $\rightarrow n/n_G$
- The Physics of the turbulence spreading phenomenon

Key Questions

- What is the anomalous physics when Ku (a local parameter) exceeds unity?
- When does the contamination/spreading process dominate over local production?
- How is the often-quoted adiabaticity parameter $\alpha = \frac{k_{\parallel}^2 v_{th,e}^2}{\nu_{ei} |\omega|}$ linked to $Ku > 1$ and the spreading phenomenon?

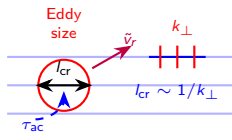
The physics

The Kubo Number Ku

Definition

$$Ku = \frac{\tilde{v}_r \cdot \tau_{ac}}{l_{cr}}, \quad (\text{R. Kubo, 1963})$$

- $\tilde{v}_r$: radial velocity fluctuation
- τ_{ac} : auto-correlation time from e-folding time of $R(\tau) = \langle \tilde{n}(t) \tilde{n}(t + \tau) \rangle / \langle \tilde{n}(t)^2 \rangle$
- $l_{cr} = 1/\sigma_k$: radial correlation length estimated from inverse of the spectral width of radial wavenumber spectrum



$Ku < 1$: Diffusive Regime $\rightarrow$
Many kicks in correlation length

$Ku > 1$: Persistent Regime $\rightarrow$
Long τ_{ac} , few kicks

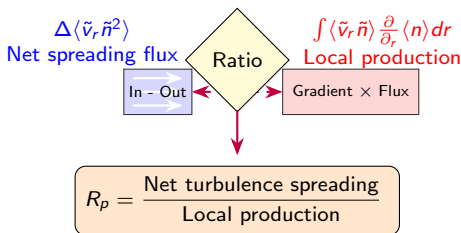
Production Ratio R_p

Definition

The production ratio is the ratio of net spreading drive to net local production in interval. R_p is defined as:

$$R_p = \frac{\Delta \langle \tilde{v}_r \tilde{n}^2 \rangle}{-\int_{r-\delta}^{r+\delta} \langle \tilde{v}_r \tilde{n} \rangle \frac{\partial \langle n \rangle}{\partial r} dr}, \quad (\text{c.f. P. Manz, T. Wu, P.D.})$$

where the interval referenced above is from LCFS to $\max(1/L_p)$.



Adiabaticity α - A Local Parameter

Definition

$$\alpha = \frac{k_{\parallel}^2 v_{\text{th},e}^2}{\nu_{ei} |\omega|}, \quad k_{\parallel} \approx \frac{1}{qR}$$

- $v_{\text{th},e} = \sqrt{K_B T_e / m_e}$: the electron thermal velocity
- ν_{ei} : electron-ion collision frequency
- $|\omega|$: frequency of the drift wave

$\alpha > 1$:

- Adiabatic electron response
- Drift wave turbulence
- Zonal flow generation

$\alpha \ll 1$:

- Non-adiabatic electron response
- Hydrodynamic turbulence
- Zonal flow collapse

Important: α linked to the density limit (Many authors)

Data & Experiment

Experimental Procedure R_p

Scans

- ① $n/n_G \rightarrow$ Greenwald fraction: Increase n/n_G by gas puffing
- ② $I_p \rightarrow$ Plasma current: Vary I_p at fixed density
- ③ $\alpha \rightarrow$ Adiabaticity: Vary α with I_p and Ku

The data are based on: J-TEXT (T. Long) and HL-2A (T. Wu)

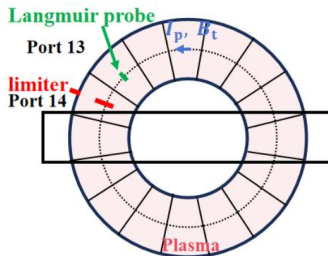
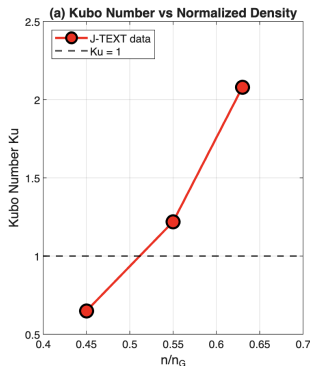


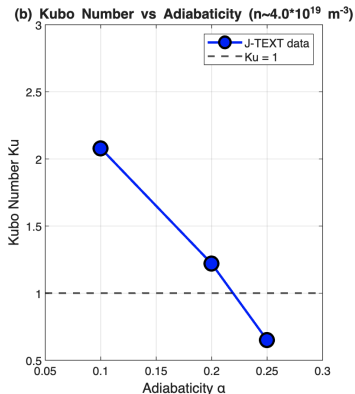
Figure: Schematic top view of J-TEXT tokamak

Experimental Results: Kubo Number Ku vs n/n_G (Global)



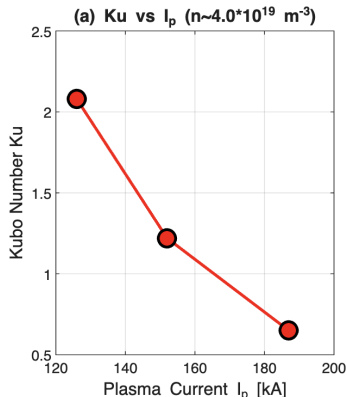
- The Kubo number Ku increases from 0.7 to 2.1 ($Ku > 1$ clear)
- The transition through $Ku = 1$ occurs near $n/n_G \sim 0.5$
- Zonal flow collapse at high n/n_G (R. Hong)

Cont'd: Ku vs α



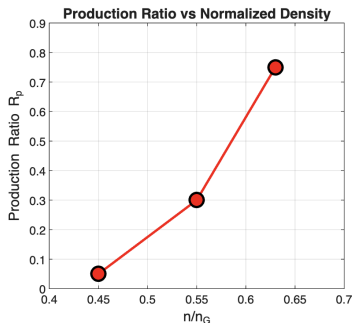
- As α decreases (increasing density due to increased electron-ion collisions), the Kubo number Ku increases
- This reveals an inverse relationship: $Ku \propto 1/\alpha$
- Higher collisionality (lower α) leads to longer auto-correlation times τ_{ac} and thus, higher Ku

Cont'd: K_u vs I_p



- Increasing the plasma current I_p reduces the Kubo number K_u
- A higher current increases the Greenwald density n_G , which lowers n/n_G at fixed density
- This result is consistent with the density scan observations

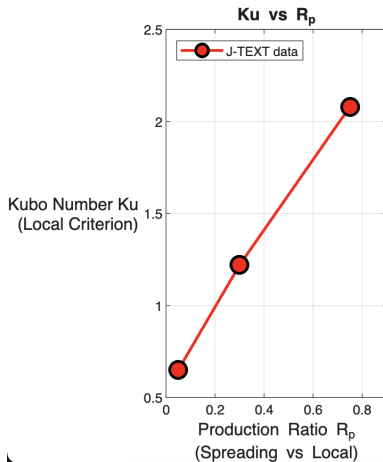
R_p (Production Ratio) vs n/n_G



- The production ratio R_p increases from 0.1 to 0.8 as n/n_G increases from 0.5 to 0.7
- At the highest density, R_p approaches unity
- This suggests a transition from a local-production-driven regime to a spreading-significant regime approaching the DL

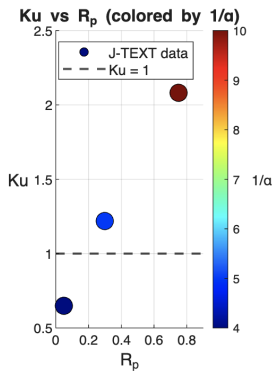
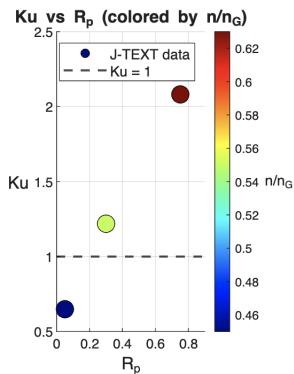
Key Results

Ku vs R_p : The Master Correlation



- Ku and R_p are strongly correlated and increase together
- **Long auto-correlation times and spreading strength are physically linked**

Ku vs R_p Colored by n/n_G and $1/\alpha$



- Color coding by n/n_G reveals the density dependence
- Higher density corresponds to higher values of both Ku and R_p

- $1/\alpha$ decreases as R_p and Ku increase
- Lower α indicates an hydrodynamic regime, favoring turbulence spreading

Partial Summary on Ku and R_p

Kubo Number Ku

- $Ku = \tilde{v}_r \tau_{ac} / l_{cr}$ distinguishes diffusive mixing ($Ku < 1$) from persistent coherent mixing ($Ku > 1$)
- Inverse relationship with adiabaticity: $Ku \propto 1/\alpha$

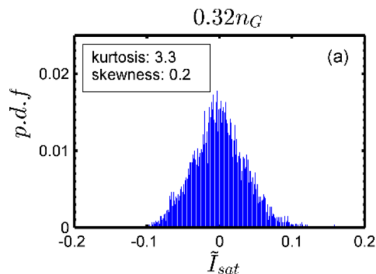
Production Ratio R_p

- As n/n_G increases from 0.5 to 0.7, R_p increases from 0.1 to 0.8
- $R_p \rightarrow 1$ suggests transition from local-production-driven to spreading-significant regime
- **Higher density $\rightarrow$ higher Ku and higher $R_p \rightarrow$ toward the operational limit n_G**

What's the Physics when Ku is high? $\rightarrow$ Blobs?

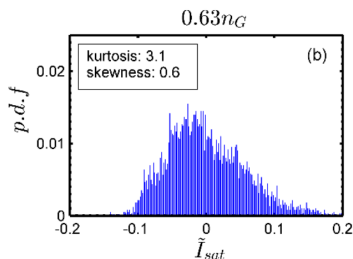
High $Ku \longleftrightarrow$ Blobs?

An Interplay of Ku and Skewness - T. Long



Key Observations

- As density approaches n_G , skewness also increases $\rightarrow$ Skewness increases with Ku
- Positive skewness indicates **blob-dominated** turbulence
- Blobs carry density radially outward



Large Ku Turbulence Spreading $\longleftrightarrow$ Blob Propagation

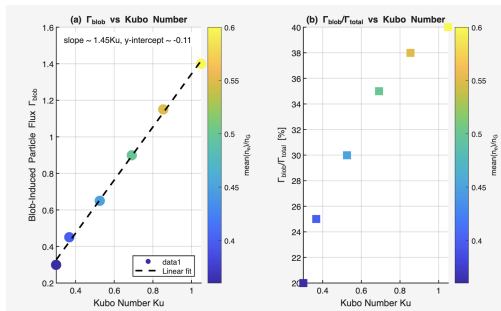
Blob Kubo Number

The blob Kubo number is defined as:

$$Ku_{\text{blob}} = \frac{v_{\text{prop}} \cdot \tau_{\text{life}}}{l_{\text{pene}}}$$

- v_{prop} : blob propagation speed
 - τ_{life} : blob lifetime
 - l_{pene} : radial blob penetration depth
-
- **At high n/n_G (low α), $Ku_{\text{blob}} \sim 1$ across all measured conditions**
 - Blobs self-organize to maintain $v_{\text{prop}} \tau_{\text{life}} \sim l_{\text{pene}}$
 - **The duality of $Ku > 1$ and $Ku_{\text{blob}} \gtrsim 1$ for large n/n_G suggests that turbulence $\longleftrightarrow$ a gas of blobs, not "modes"**

Blob Fraction of Flux Increases with Kubo Number



- The blob-induced particle flux Γ_{blob} is calculated using conditional averaging
- The fraction $\Gamma_{\text{blob}}/\Gamma_{\text{total}}$ increases with Ku , from approximately 25% at $Ku = 0.40$ to approximately 60% at $Ku = 0.60$
- As $Ku \rightarrow 1$, we anticipate $\Gamma_{\text{blob}}/\Gamma_{\text{total}}$ to exceed 80%

Conclusions

- 1 **The $Ku \sim R_p$ correlation demonstrates that spreading and contamination/spreading processes underlie $Ku > 1$ phenomena.**
The production ratio tracks the transition from local-production-dominated to interaction-dominated turbulence as we approach the density limit
- 2 **Blob spreading** emerges with $Ku_{\text{blob}} \rightarrow 1$ at high n/n_G . The blob-induced particle flux exceeds 50% of total transport when $Ku > 0.5$, confirming that blobs become the dominant transport channel in the interaction-dominated regime
- 3 The adiabaticity α is a key **local parameter** in determining Ku and R_p
- 4 **Message for modelers:** At high n/n_G , turbulence is best modeled as a gas of blobs, not as an ensemble of linear modes. Traditional quasi-linear models fail!

Future Aims

- 1 **Extend to larger devices:** Apply Ku and R_p analysis to HL-3 and DIII-D to verify scaling relationships across machine sizes
- 2 **Quantify the R_p phase space:** Map the boundary between local-production-dominated ($R_p < 1$) and spreading-dominated ($R_p > 1$) regimes as a function of n/n_G and I_p
- 3 **Analyze the production ratio for gyrokinetic simulations.** (See T. Long, M.J. Choi, and P.H. Diamond, Philos T Roy Soc A, in press, 2026)
- 4 **Study edge-core coupling:** Investigate how turbulence spreading from the edge contaminates the core and whether core density saturation via avalanching is linked to $R_p > 1$ edge dynamics
- 5 **Study the relation between R_p and zonal flow:** R_p vs Shearing Kubo number
- 6 **Revise the production ratio without blob contributions**

Thank You!

Production Ratio R_p Continued (BACKUP)

Derivation

The production ratio emerges naturally from the turbulence intensity evolution equation:

$$\frac{\partial}{\partial t} \left(\frac{\langle \tilde{n}^2 \rangle}{2} \right) = -\langle \tilde{v}_r \tilde{n} \rangle \frac{\partial \langle n \rangle}{\partial r} - \frac{\partial}{\partial r} \left(\frac{\langle \tilde{v}_r \tilde{n}^2 \rangle}{2} \right) - \epsilon$$

where the first term on the right is local production, the second term is the turbulence spreading flux, and the third term is the external dissipation sources. R_p is defined as:

$$R_p = \frac{\Delta \langle \tilde{v}_r \tilde{n}^2 \rangle}{-\int_{r-\delta}^{r+\delta} \langle \tilde{v}_r \tilde{n} \rangle \frac{\partial \langle n \rangle}{\partial r} dr}$$

The Contamination Analogy (Pomeau, 1986) (BACKUP)



"Each oscillator if in a turbulent state may either relax spontaneously towards its quiescent state or contaminate its neighbors. - Pomeau, 1986"

Pomeau's Perspectives

- The analysis of reaction-diffusion systems reveals that the interaction between nearby oscillators may be seen as a sort of contamination process
- This is precisely where the turbulent state spreads through interactions rather than local instabilities, analogous to the $R_p < 1$ regime
- The robustness of the spreading front is consistent with the interaction-driven regime becoming dominant near operational limits